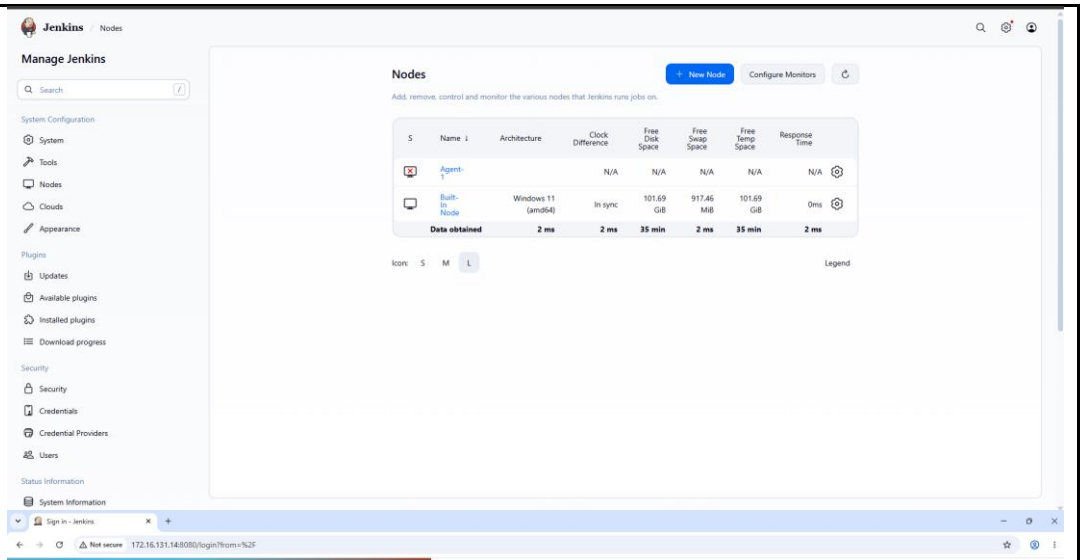
Semester	B.E. Semester VII – Computer Engineering
Subject	DevOps Lab
Subject In-charge	Prof. Snehal Andhere

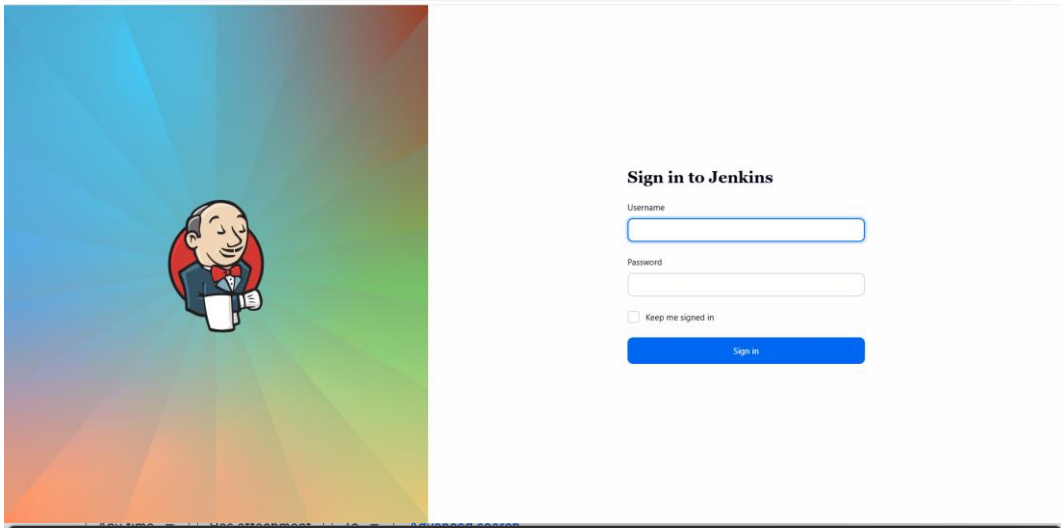
Student Name	Mayuresh Sawant
Roll Number	23102A0029
Grade and Subject Teacher's Signature	

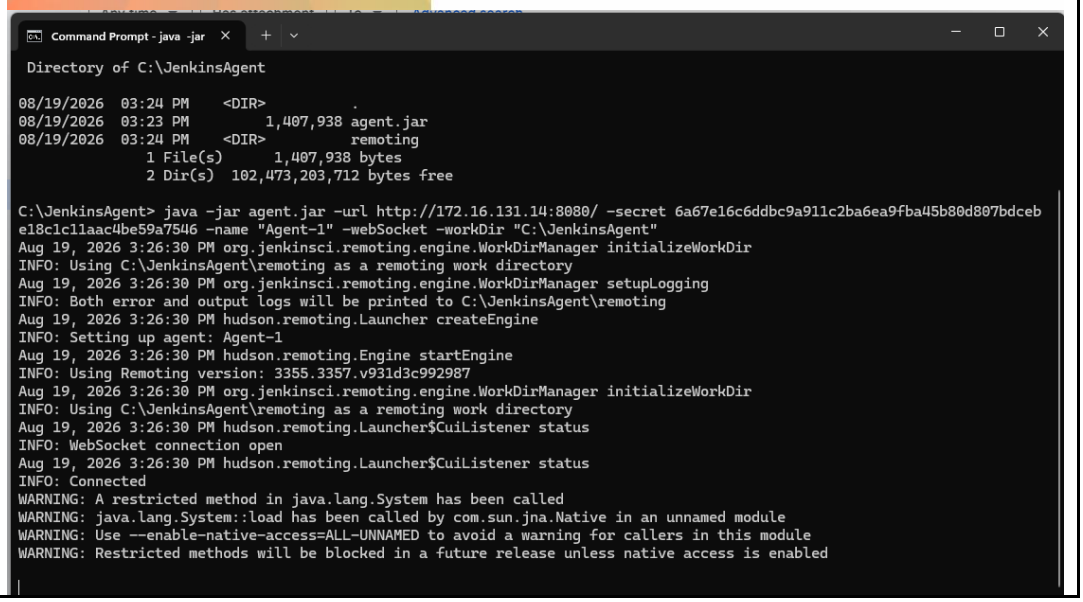
Experiment Number	05	
Experiment Title	Jenkins Application – Master Slave Architecture	
Resources / Apparatus Required	Hardware:	Software: Jenkins,Github
Description	To understand Jenkins Master-Slave Architecture and scale your Jenkins standalone implementation by implementing slave nodes.	
Program & Output	Step 1: Check Jenkins on PC 1 On PC 1, start Jenkins. Open the browser: http://localhost:8080 You should see the Jenkins dashboard. So now: PC 1 → Jenkins Controller Step 2: Find the IP address of PC 1 Open Command Prompt on PC 1. Type: ipconfig You may see something like: IPv4 Address: 192.168.1.10 Your Jenkins address for other computers becomes: http://192.168.1.10:8080 Step 3: Test from PC 2 Go to PC 2. Open Chrome or Edge. Enter: http://192.168.1.10:8080 If Jenkins opens, excellent. It means: PC 1 <--------> PC 2 can communicate. This step is very important. Step 4: Check Java on PC 2 Open Command Prompt on PC 2.	

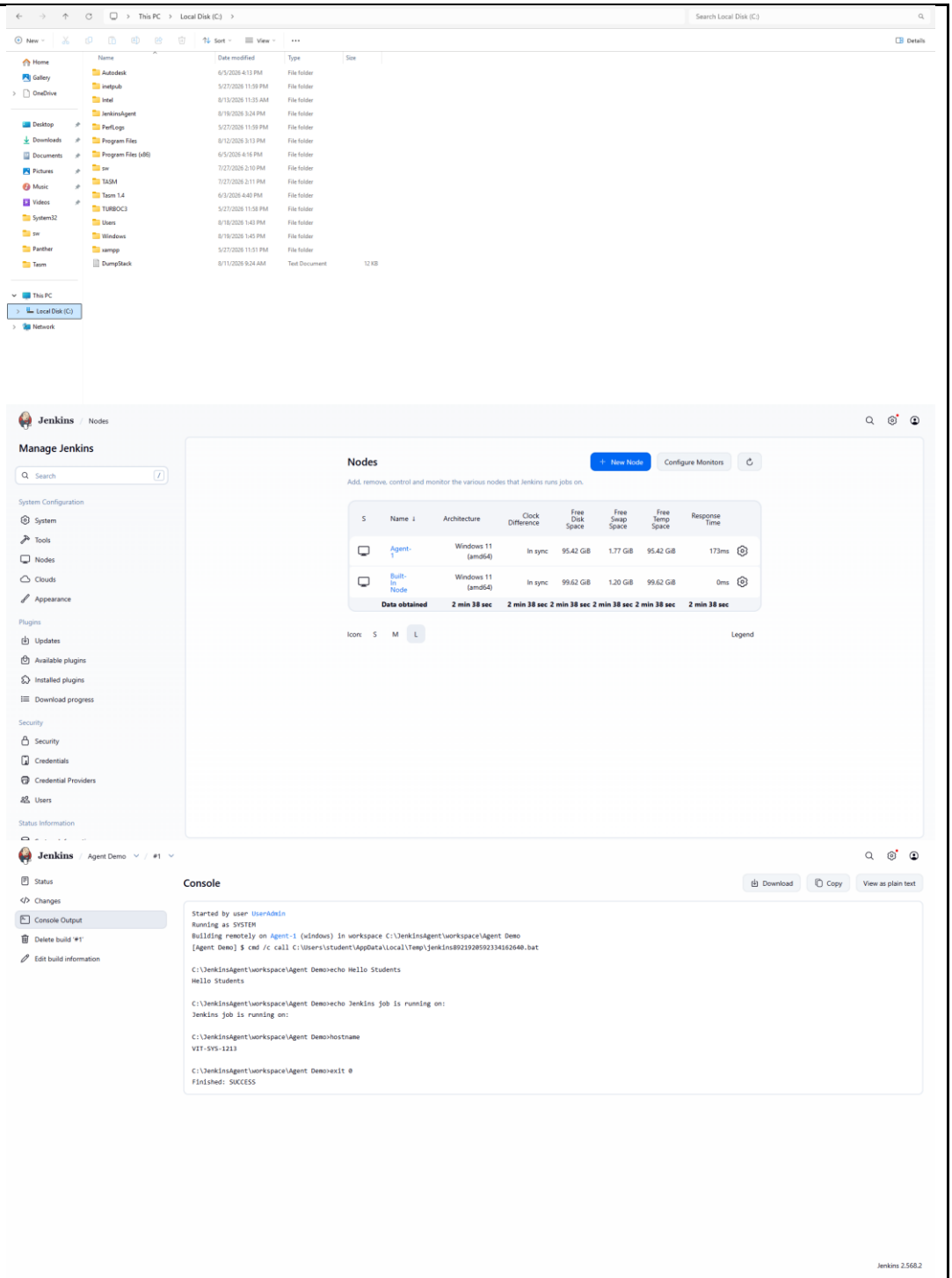
	Run: java -version If Java is installed, you will see the version. Something like: java version "21..." Java is needed because the Jenkins Agent program runs using Java. Step 5: Create an Agent in Jenkins Now go back to Jenkins on PC 1. Open: Jenkins Dashboard → Manage Jenkins → Nodes → New Node Enter the name: Agent-1 Select: Permanent Agent Click Create. Now Jenkins knows: "I want to connect another computer called Agent-1." Step 6: Give Agent information You will see several options. For beginners, use something like: Number of executors: 1 This means: Agent-1 can execute one Jenkins job at a time. For Remote Root Directory enter: C:\JenkinsAgent This is simply the folder where Jenkins will store its files on PC 2. For Label enter: windows Label is like giving the computer a category. Later Jenkins can say: "Run this job only on the Windows machine." Step 7: Create the folder on PC 2 On PC 2 create: C:\JenkinsAgent You can do it manually or in Command Prompt: mkdir C:\JenkinsAgent Then: cd C:\JenkinsAgent Step 8: Connect Agent-1 After saving your Agent configuration, Jenkins gives you a command. It looks like: java -jar agent.jar -url http://192.168.1.10:8080/ -secret XXXXX -name "Agent-1" It simply means: Java → Start Jenkins Agent program → Connect to Jenkins Controller → Tell Jenkins: "I am Agent-1" You run this command on PC 2. After successful connection, Jenkins should show:

	Agent-1 Online Now the architecture is ready. PC 1 PC 2 Jenkins Controller → Jenkins Agent-1 Create a simple Jenkins job. Tell Jenkins: "Do not execute this job on the Controller. Execute it on Agent-1." Create: Dashboard → New Item → Freestyle Project Name it: AgentDemo Inside configuration, find: Restrict where this project can be run Enter the label: windows Why? Because Agent-1 has label: windows So, Jenkins understands: Job requires windows → Which computer has windows label? → Agent-1→ Send job to Agent-1 Add a very simple command Add: Build Steps → Execute Windows batch command Enter: echo Hello Students echo Jenkins job is running on: hostname Save it. Click: → Build Now Then open: → Console Output You may see: Hello Students Jenkins job is running on: LAB-PC-22 If LAB-PC-22 is PC 2, congratulations. Your Controller has successfully sent the job to another computer


Jenkins
Nodes
Manage Jenkins
Search
System Configuration
System
Tools
Nodes
Clouds
Appearance
Plugins
Updates
Available plugins
Installed plugins
Download progress
Security
Security
Credentials
Credential Providers
Users
Status Information
System Information
Nodes
New Node
Configure Monitors
Add, remove, control and monitor the various nodes that Jenkins runs jobs on.
S
Name
Architecture
Clock Difference
Free Disk Space
Free Swap Space
Free Temp Space
Response Time
Agent-1
N/A
Built-In Node
Windows 11 (amd64)
In sync
101.69 GiB
917.46 MiB
101.69 GiB
0ms
Data obtained
2 ms
35 min
Icon: S M L
Legend


Sign in - Jenkins
Not secure
172.16.131.14:8080/login?from=%2F
Sign in to Jenkins
Username
Password
Keep me signed in
Sign in


Command Prompt - java -jar
Directory of C:\JenkinsAgent
08/19/2026 03:24 PM <DIR> .
08/19/2026 03:23 PM 1,407,938 agent.jar
08/19/2026 03:24 PM <DIR> remoting
1 File(s) 1,407,938 bytes
2 Dir(s) 102,473,203,712 bytes free
C:\JenkinsAgent> java -jar agent.jar -url http://172.16.131.14:8080/ -secret 6a67e16c6ddbc9a911c2ba6ea9fba45b80d807bdceb
e18c1c11aac4be59a7546 -name "Agent-1" -webSocket -workDir "C:\JenkinsAgent"
Aug 19, 2026 3:26:30 PM org.jenkinsci.remoting.engine.WorkDirManager initializeWorkDir
INFO: Using C:\JenkinsAgent\remoting as a remoting work directory
Aug 19, 2026 3:26:30 PM org.jenkinsci.remoting.engine.WorkDirManager setupLogging
INFO: Both error and output logs will be printed to C:\JenkinsAgent\remoting
Aug 19, 2026 3:26:30 PM hudson.remoting.Launcher createEngine
INFO: Setting up agent: Agent-1
Aug 19, 2026 3:26:30 PM hudson.remoting.Engine startEngine
INFO: Using Remoting version: 3355.3357.v931d3c992987
Aug 19, 2026 3:26:30 PM org.jenkinsci.remoting.engine.WorkDirManager initializeWorkDir
INFO: Using C:\JenkinsAgent\remoting as a remoting work directory
Aug 19, 2026 3:26:30 PM hudson.remoting.Launcher$CuiListener status
INFO: WebSocket connection open
Aug 19, 2026 3:26:30 PM hudson.remoting.Launcher$CuiListener status
INFO: Connected
WARNING: A restricted method in java.lang.System has been called
WARNING: java.lang.System::load has been called by com.sun.jna.Native in an unnamed module
WARNING: Use --enable-native-access=ALL-UNNAMED to avoid a warning for callers in this module
WARNING: Restricted methods will be blocked in a future release unless native access is enabled



Conclusion:	The experiment successfully implemented Jenkins Controller-Agent architecture. PC 1 acted as the Controller, while PC 2 was configured as Agent-1. A Jenkins job was successfully executed on Agent-1, demonstrating job distribution and scalability in Jenkins.